Ultrasound-Guided Closed-Loop Control of Magnetic hydrogel robotics with Autonomous Switching Modes

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Introduction

Inspired by aquatic organisms, soft hydrogel microrobots have recently emerged as attractive candidates for minimally invasive biomedical interventions due to their flexibility, biocompatibility and capability to navigate intricate biological environments.Effective navigation within such environments inherently requires precise real-time control with accurate localization and continuous positional feedback. Among various propulsion methods, magnetic actuation is particularly promising due to its non-invasive nature, remote operability, and strong clinical compatibility. To enable precise magnetic control in vivo, appropriate imaging techniques are essential; ultrasound imaging has become the most common choice due to its unique combination of real-time capability, non-invasiveness, deep-tissue penetration, high spatial resolution, and cost-effectiveness.

However, while magnetic propulsion combined with ultrasound imaging has successfully facilitated precise control of rigid micro-robotics, extending these strategies to soft, deformable hydrogel robots remains challenging. Unlike rigid micro-robotics, whose movements can be reliably controlled by straightforward magnetic gradient forces, soft hydrogel robots exhibit nonlinear and time-dependent motion behaviors due to their structural deformation and complex interactions with surrounding fluid. For instance, jellyfish-inspired soft robots encounter fluid resistance that counteracts their intended shape changes, necessitating adaptive and robust control strategies that dynamically respond to hydrodynamic forces and structural deformation. To date, fully autonomous navigation of hydrogel-based robots using ultrasound-guided closed-loop control remains unrealized. Moreover, utilizing closed-loop control systems to intelligently switch between multiple swimming modes has been minimally explored. Achieving such dynamic behavioral adaptability would represent a transformative advancement, significantly narrowing the gap between artificial robotic systems and the natural agility observed in biological organisms.

In this study, we propose a novel ultrasound-guided closed-loop control approach specifically designed to address these critical limitations. Inspired by the unique locomotion of crown-of-thorns starfish (COTS), we developed magnetic-driven hydrogel microrobots using digital light processing (DLP)-based 3D printing, embedding magnetic nanoparticles within poly(ethylene glycol) diacrylate (PEGDA)-based hydrogels. Under precisely modulated magnetic fields, the robots autonomously transition between two distinct locomotion modes: omnidirectional swimming, characterized by rhythmic oscillations under sawtooth waveforms, and spherical rolling motion induced by sinusoidal waveforms with phase offsets. To achieve robust and precise real-time control, we combined ultrasound-based position tracking with a customized inter-frame difference algorithm enhanced by adaptive Kalman filtering, effectively accounting for measurement and process noise. Furthermore, we developed a sophisticated control architecture integrating model predictive control (MPC) and PID algorithms to accurately track planned trajectories, dynamically switching between locomotion modes as required. Our experimental validation in biomimetic phantoms demonstrated precise autonomous navigation, successful obstacle traversal, and reliable mode switching capabilities, achieving localization accuracy within 2 mm.

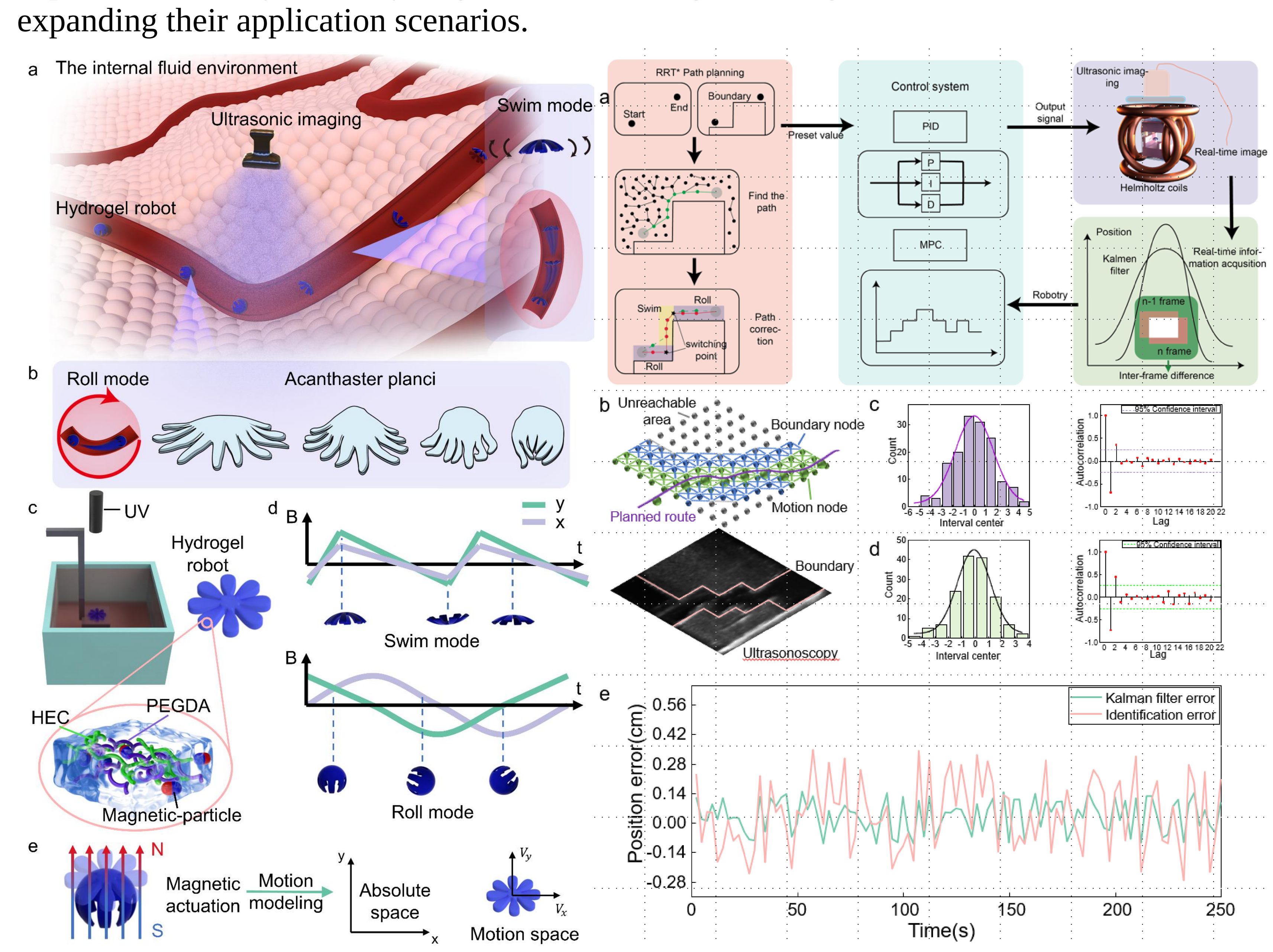
Methods

To assess the feasibility and efficacy of closed-loop ultrasound-guided control in soft hydrogel robots, we fabricated a biomimetic robot inspired by the crown-of-thorns starfish (COTS) using digital light processing (DLP)-based 3D printing. The robot was synthesized from poly(ethylene glycol) diacrylate (PEGDA) monomers, acrylamide (AAM) crosslinkers, and embedded magnetic nanoparticles, followed by magnetization to ensure responsiveness to external magnetic fields. And after magnetization, the spatial positions of the hydrogel robots were marked in the absolute space, and the speeds of the hydrogel robots were analyzed in the relative space.

We obtain the spatial position of the robot through the inter-frame difference method, and use the Kalman filter to improve the recognition accuracy. The spatial path was obtained through the RRT* path planning algorithm. A control framework combining PID controller and MPC controller was designed, which enables the water gel robot to autonomously switch between swimming and rolling motion modes during autonomous navigation..

Results

We used the fabricated mimics to simulate the in-vivo application environment of the hydrogel robots. The slope-passing ability of the hydrogel robot was investigated, and the different application conditions for the swimming mode and rolling mode of the hydrogel robot were demonstrated. Under our control system, the hydrogel robots are capable of autonomously navigating to the target location. Furthermore, we also explored the ability of the hydrogel robots to navigate through unknown terrains, further expanding their application scenarios.



Conclusions

In summary, we report a method for constructing a real-time navigation control system for a hydrogel robot with multiple motion modes based on ultrasound imaging. We have achieved the autonomous switching of motion modes by the hydrogel robot during movement to navigate more efficiently and safely within the body. The control system effectively ensures the stability and accuracy of the hydrogel robot's navigation within the body. Additionally, we explored the situation where obstacles cannot be detected by ultrasound. The control system enables the hydrogel robot to complete global navigation by detecting the contours of the obstacles. Our method demonstrates the potential of hydrogel robots in clinical applications and intelligent control, providing a strategy for precise control of soft robots under ultrasound imaging.

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